Causal-ARD: Zero-shot Reward Design for Automated Skill Generation

***Abstract*—Automated Reward Design (ARD) aims to replace manual reward engineering in reinforcement learning with language-driven reward function synthesis. However, existing approaches based on large language models (LLMs) often rely on semantic associations and iterative environmental feedback to refine task-specific reward hypotheses. This paradigm not only results in inefficient reasoning but also makes LLMs susceptible to semantically plausible yet causally spurious reward components, leading to ineffective optimization. To address these limitations, we propose Causal-ARD, a zero-shot ARD framework that learns transferable causal guidance from offline multi-task interaction data and uses it to constrain reward construction for unseen tasks. Causal-ARD first estimates a coarse reward causal prior from controlled reward-component interventions, and then introduces a joint optimization module that integrates Explicit Mechanism Decoupling with Confidence-Aware Soft Fusion to refine this prior using micro-level trajectories. The resulting prior provides reward-relevant structural guidance while accounting for state-dependent variations in the offline trajectories. During inference, Causal-ARD serializes the refined reward causal prior into explicit constraints that guide an LLM to select supported reward components, prune terms unsupported by the supplied prior, and synthesize executable reward code in a single pass. Our work establishes an interpretable, causal-prior-guided formulation of the ARD problem. Extensive experiments on complex continuous control benchmarks demonstrate that Causal-ARD generates effective rewards without feedback-driven refinement, avoids target-task evolutionary search, achieves performance comparable to the evaluated iterative methods, and generalizes across unseen tasks and diverse robotic embodiments.**

***Index Terms*—Automated Reward Design, Causal Prior, Zero-Shot Reward Generation, Reinforcement Learning**

I. Introduction

ENDOWING robots with the capability for autonomous skill acquisition remains a longstanding objective within the field of embodied artificial intelligence. In this context, the design of reward functions has always been a challenging issue in the automatic generation of skills based on reinforcement learning (RL) [1], [2]. Recently, LLMs have given rise to the emergence of Automated Reward Design (ARD) paradigms, demonstrating significant potential in synthesizing executable reward functions through semantic reasoning [3], [4]. Given a task description, these methods prompt an LLM to generate candidate reward components and iteratively refine them through environmental feedback until a satisfactory reward function is obtained. By utilizing the extensive knowledge and reasoning capabilities of LLMs, these approaches mitigate the reliance on human expertise in reward engineering [5], [6].

However, current LLM-based reward design methods predominantly rely on trial-and-error to select candidate reward functions. During the optimization process, LLMs identify factors potentially correlated with the task as candidate reward components, yet such candidate components may exhibit spurious correlations with the task objectives [7], [8]. For example, in the *ShadowHandLiftUnderarm* task (Fig. 1(a)), the agent aims to learn the skill of grasping and lifting a pot with both hands. An unguided LLM may introduce `velocity`, i.e., the instantaneous lifting speed of the pot, as a reward component. Although this term is semantically related to "lifting", it provides a high-frequency but causally ambiguous signal during continuous physical interaction, which may provide noisy learning signals; the illustrated run exhibits Optimization Collapse. In the *ShadowHandScissors* task (Fig. 1(b)), the LLM treats `scissors_open` as a reward component. However, the agent exploits this signal by maintaining the scissors slightly open, allowing it to accumulate rewards without achieving the genuine goal. This is actually a form of Specification Gaming [9], [10]. These examples show that semantically plausible reward components can be misaligned with the task objective. In the absence of reliable prior knowledge to guide the search, certain approaches resort to evolutionary strategies [3], i.e., utilizing "trial-evaluation-mutation-feedback" loops to filter out spuriously correlated factors. However, the inherent limitations of LLMs in causal attribution and counterfactual reasoning often undermine the efficacy of these methods, preventing consistent performance gains [11].

More importantly, due to the underutilization of inter-task reward relationships, existing methodologies are constrained to designing and optimizing reward functions on a case-by-case basis. Indeed, within identical dynamical systems, reward components across distinct tasks often exhibit shared causal structures [12], [13]. Zhang et al. [14] study identifiability of reward functions and causal structure under specific modeling assumptions. These results motivate reusable structural guidance for reward design; they do not establish identifiability for our reward-weight intervention protocol. Consequently, an interesting question is raised: *Can transferable reward causal priors be learned from offline multi-task data and used to construct rewards for unseen tasks without evolutionary refinement?*

In this paper, we propose **Causal-ARD**, a zero-shot automated reward design framework that learns reusable causal guidance from offline multi-task interaction data. Its central objective is to learn reusable reward-relevant relations among candidate components and physical variables [12], and then use these relationships to constrain reward construction for unseen tasks. The learned structure serves as a *reward causal prior*: it encodes candidate reward-relevant dependencies, guides component pruning, and provides an explicit basis for selecting and organizing reward terms. At inference time, Causal-ARD converts this prior into structured constraints for an LLM, enabling the LLM to generate executable rewards without evolutionary search or task-specific environmental feedback. Because the prior is shared across tasks whose

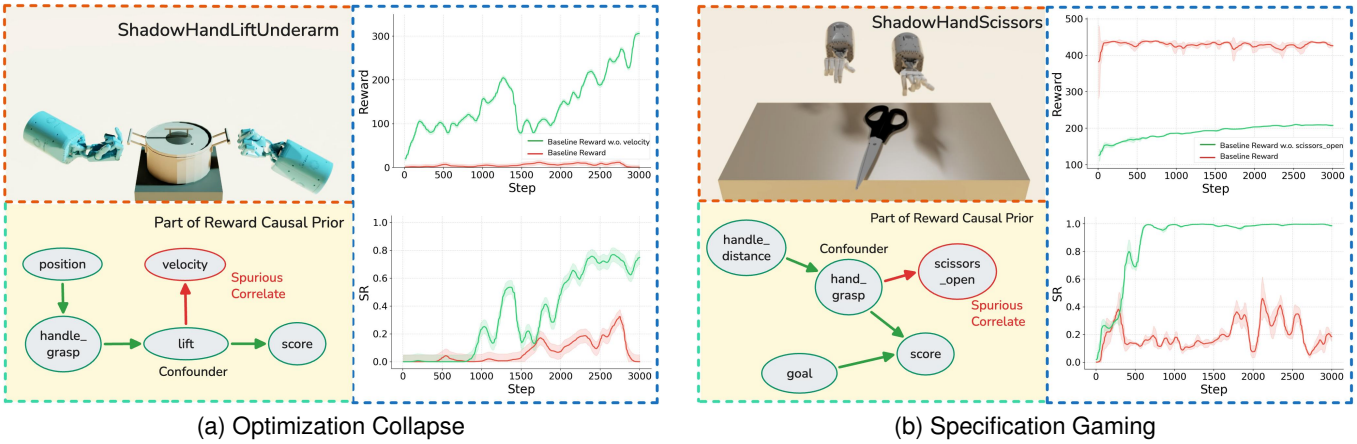


Fig. 1. **Reward misalignment in LLM-driven Automated Reward Design.** The bottom-left panels illustrate reward-component causal graphs, where green nodes denote retained components and red nodes denote components excluded by the learned prior. The right panels compare reward and Success Rate (SR) curves between baseline rewards with spurious correlates (red) and causally pruned rewards (green). (a) Optimization Collapse in *ShadowHandLiftUnderarm*. (b) Specification Gaming in *ShadowHandScissors*.

reward variables are covered by the learned structural vocabulary, it can be reused to reduce feedback-driven reward search for new skills.

Causal-ARD learns the reward causal prior through a coarse-to-fine procedure. We first estimate a coarse structural prior from controlled reward-component interventions using a causal foundation model, and subsequently refine it with micro-level trajectories of structural variables [15], [16]. To robustly extract the causal skeleton from offline trajectories of structural variables, we further propose the Explicit Mechanism Decoupling (EMD) with Confidence-Aware Soft Fusion, which together constitute a joint optimization module. By integrating evidence across multiple tasks, this module accounts for state-dependent transient physical interactions during reconstruction and corrects structural hallucinations in the coarse prior, yielding a refined reward causal prior. For a new task, Causal-ARD serializes the refined structure into a causal topology prompt, filters components that are unavailable in the target observation space, and guides the LLM to prune spurious reward terms before generating executable code [17].

Extensive experiments across complex continuous control tasks demonstrate that our approach improves reward alignment and task performance in the evaluated settings. Remarkably, our approach achieves zero-shot reward generation by generating executable reward functions for unseen tasks in a single pass without feedback-driven reward refinement. This capability bypasses the computationally expensive evolutionary trial-and-error loops, while yielding task performance that rivals state-of-the-art results.

In summary, our main contributions are as follows:

1) We propose Causal-ARD, an interpretable framework that uses offline reward causal priors to generate executable reward functions for unseen tasks without feedback-driven refinement.
2) We introduce a coarse-to-fine reward causal prior learning method. The proposed EMD and Confidence-Aware Soft Fusion use micro-level trajectories to model state-dependent variations and calibrate a reusable reward-relevant structure.
3) The proposed causal-prior-guided reward construction reduces feedback-driven reward search on new tasks while improving zero-shot performance. Experiments show that our zero-shot method matches state-of-the-art iteration-based ARD methods on unseen Dexterity tasks and transfers across different robotic embodiments and environments. Codes will be available after peer review on the project website.

II. Related Work

Automated Reward Design. Automated Reward Design (ARD) remains a central challenge in reinforcement learning-based skill acquisition. Traditional approaches often rely on hand-crafted reward shaping [18], [19] or inverse reinforcement learning from expert demonstrations [20], which require substantial human expertise or high-quality data. Recent LLM-based methods have advanced ARD by prompting LLMs to generate executable reward functions from task descriptions [3], [4], [21]. Although these methods reduce manual engineering, they generally rely on iterative environmental feedback to evaluate, mutate, and refine task-specific reward candidates [3], [22], [23]. Moreover, LLMs often infer reward terms from semantic associations rather than causal relevance, which can introduce plausible but ineffective components and lead to Optimization Collapse or Specification Gaming [24]. Causal-ARD instead uses causal knowledge learned from offline source tasks to constrain reward construction, thereby targeting zero-shot reward design without feedback-driven refinement on the target task.

Causal Priors for Reward Design. Causal discovery provides directed structure for reasoning beyond observational correlation, and continuous optimization has enabled causal graph learning from data [25], [26]. In high-dimensional physical systems, however, dense interactions, incomplete observations, and limited interventions make purely data-driven structure estimation unreliable [27], [28]. Prior-guided

approaches reduce this ambiguity by incorporating temporal assumptions through dynamic Bayesian networks [29] or semantic knowledge from LLMs [8], [30], but such priors may contain biased effects and structural hallucinations [8], [11], [30]. Static causal graphs may also conflate persistent relations with transient physical interactions such as contact and friction changes [25], [27], [29]. Related studies have shown that causal representations can support policy generalization and model alignment by separating intervention-relevant mechanisms from spurious correlations [31]–[33]. Building on these ideas, Causal-ARD learns a reusable reward causal prior from controlled reward-component interventions and physical trajectories, refines unreliable prior relations using state transitions, and converts the resulting structure into explicit constraints for zero-shot reward construction.

III. Preliminaries

In this section, we introduce the key notations used throughout this work. We first introduce Structural Causal Models (SCMs), and then formulate the Automated Reward Design problem.

Structural Causal Models. To formalize causal relationships among reward-relevant variables, we adopt the framework of Structural Causal Models (SCMs) [34]. We assume that the task-relevant physical quantities are represented by a set of d structural variables $\mathcal{V} = \{v^1, v^2, \ldots, v^d\}$, where each variable corresponds to a physically meaningful reward component, such as object position, velocity, or contact force. An SCM represents directed dependencies using an adjacency matrix $M \in \mathbb{R}^{d\times d}$, where a non-zero entry $M^{i,j}$ indicates a direct causal edge from v^j to v^i. The temporal evolution of each variable is described by:

$$v^i_{t+1} := \psi_i(Pa(v^i)_t, U^i_t), \tag{1}$$

where $\mathcal{V}_t$ denotes the variables at time t, and $Pa(v^i)_t \subseteq \mathcal{V}_t$ denotes the causal parents of v^i identified by $\{v^j \mid M^{i,j} \neq 0\}$. Here, U^i_t denotes unmodeled exogenous influences. This SCM notation supplies causal terminology; the learned reward prior below is a structural abstraction, not an identified version of this full transition model.

Automated Reward Design. We consider an episodic environment governed by a Markov Decision Process (MDP) $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T} \rangle$, where $\mathcal{S}$ denotes the continuous state space, $\mathcal{A}$ denotes the action space, and $\mathcal{T}(s_{t+1}|s_t, a_t)$ denotes the transition dynamics. We formulate ARD as $\mathcal{P} = \langle \mathcal{M}, \mathcal{R}, \pi_{\mathcal{M}}, \mathcal{F} \rangle$. Here, $\mathcal{R}$ denotes the space of permissible parameterized reward functions, and $\pi_{\mathcal{M}} : \mathcal{R} \to \Pi$ represents an RL algorithm that optimizes a reward $r \in \mathcal{R}$ and outputs a policy $\pi \in \Pi$. The function $\mathcal{F} : \Pi \to \mathbb{R}$ denotes the non-differentiable fitness score, such as the objective task success rate. The goal of modern LLM-based ARD [3], [22] is to generate a reward function r^* that maximizes the fitness score of the policy trained under this reward:

$$r^* = \arg\max_{r\in\mathcal{R}} \mathcal{F}(\pi_{\mathcal{M}}(r)). \tag{2}$$

Since evaluating F requires costly environment interactions, existing ARD methods usually rely on iterative trial-and-error to evaluate and refine candidate rewards [3], [22]. We instead study *zero-shot ARD*, where the reward for a target task is generated without using target-environment evaluations to revise the reward. To support this setting, we learn a reward causal prior $G_R = (\mathcal{V}, \mathcal{E}_R)$ from offline source-task data, where $\mathcal{E}_R$ contains candidate reward-relevant relations informed by reward-weight interventions and physical trajectories. Given a target-task description, its available observation variables, and G_R, the objective is to construct an executable reward r^* while excluding unavailable components and terms unsupported by the supplied prior.

IV. Method

The goal of Causal-ARD is to construct executable rewards for unseen tasks without task-specific feedback-driven refinement. As illustrated in Fig. 2, the framework first learns a reusable reward causal prior from offline multi-task data and then translates this prior into explicit constraints for zero-shot LLM reward construction.

Given the structural variables $\mathcal{V}$ defined in Section III, the physical-variable part of the reward causal prior is represented by a weighted skeleton M^*_{inv} learned from offline interaction data. The skeleton encodes candidate reward-relevant relations among physical variables; it is a structured prior for reward design, rather than an identified model of the complete physical dynamics. During prior learning, we optimize M_{inv} over $\mathcal{V}$ and introduce a state-dependent transient component, later parameterized as $M_{trans}(s_{t-1};\theta)$, to account for short-lived interactions in micro-level trajectories, such as brief contacts or collisions. After convergence, M^*_{inv} is retained for constructing R^*. Macro-level component-to-fitness links $\mathcal{E}_F$ provide a separate task-outcome reference for the prompt's `score` node; score is not a coordinate of s_t and is not reconstructed by the micro-level loss.

The framework consists of three synergistic phases. First, Section IV-A extracts an initial **coarse structural prior** from macro-level interventional data using a pre-trained structured-data foundation model. Second, Section IV-B introduces a Joint Optimization Module that integrates **Explicit Mechanism Decoupling (EMD)** with **Confidence-Aware Soft Fusion** to refine the coarse prior with dense micro-level trajectories, producing M^*_{inv}. Finally, Section IV-C serializes the refined prior into a causal topology prompt, grounds it in the target task's available observations, and guides the LLM to perform explicit Causal Pruning during code generation.

A. Structural Prior Extraction

We first obtain a coarse structural prior as the initialization for reward causal prior learning. This stage addresses the difficulty of learning a reward-relevant causal skeleton directly from high-dimensional physical trajectories, where dense continuous variables, limited interventions, and redundant observations can lead to spurious causal discovery. Moreover, a task-relevant structural prior cannot be inferred from variable names or semantic associations alone; it requires interventional evidence showing how perturbations of primitive reward components affect final task performance. This

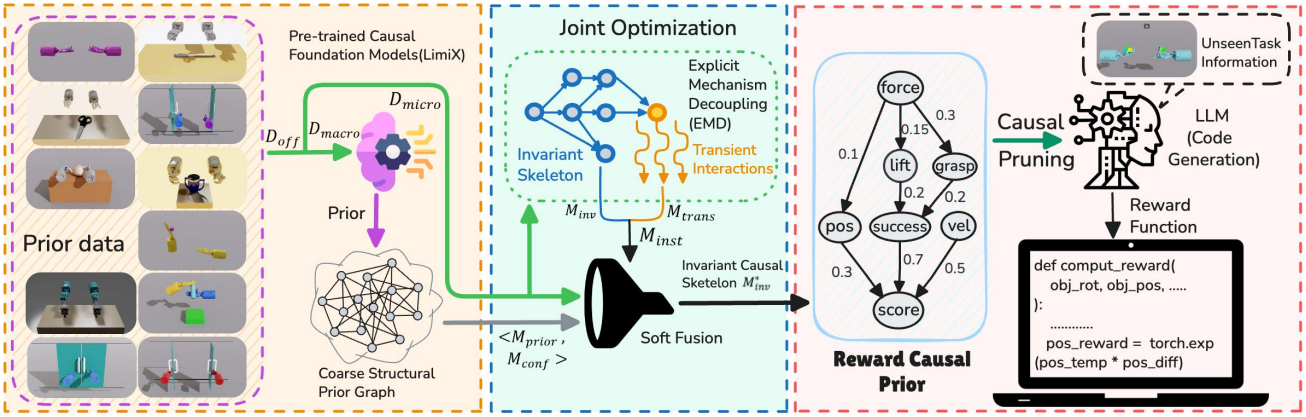


Fig. 2. **Overview of the Causal-ARD framework.** The pipeline consists of three phases: (1) **Structural Prior Extraction:** The offline interventional dataset ($\mathcal{D}_{off}$) is split into macroscopic ($\mathcal{D}_{macro}$) and microscopic ($\mathcal{D}_{micro}$) streams. A pre-trained causal foundation model (LimiX) processes $\mathcal{D}_{macro}$ to extract the initial, coarse structural prior $\langle M_{prior}, M_{conf} \rangle$. (2) **Joint Optimization Module:** The EMD module constructs the state-dependent coefficient matrix M_{inst} by combining a learnable causal skeleton M_{inv} (blue stream) with state-dependent transient physical interactions M_{trans} (orange stream). Subsequently, the Joint Optimization Module uses $\mathcal{D}_{micro}$ and Confidence-Aware Soft Fusion to account for state-dependent transient interactions and produce the refined reward causal prior M^*_{inv}. (3) **Causal-Constrained Reward Construction:** The refined prior is combined with unseen-task information and available observation variables. Through explicit Causal Pruning, the LLM generates an executable reward function R^* in a single pass.

motivates a hierarchical data collection pipeline that records both macro-level reward-component interventions and micro-level structural-variable trajectories.

Hierarchical Data Collection. To provide the causal foundation model with interventional data for extracting a task-relevant structural prior, we construct the offline dataset $\mathcal{D}_{off}$ (Fig. 3). Instead of randomly exploring the raw state space, we use ARD frameworks (Eureka and URDP [3], [22]) to extract $k \in [3, 8]$ physically meaningful reward components $\{r_1, \ldots, r_k\}$, such as target distance, joint velocity, and contact force. For cross-task causal learning, we normalize semantically equivalent task-specific components into d shared structural variables $\mathcal{V} = \{v^1, \ldots, v^d\}$, such as object position, velocity, and contact force, through the *Task-Specific Name Normalization* and *Global Atomic Reward Component Pool* (see Supplementary Material, hereafter Suppl., Sec. S-B-B). For each reward specification, we uniformly sample N weight vectors $\boldsymbol{\omega}^{(n)}$ (Suppl. Sec. S-B-C) to probe their effects on task performance. These interventions act on reward weights [35]: they change the trained policy and hence its state distribution and fitness. They are not direct interventions on physical variables v^i.

For each reward specification, the two data levels are: 1) $\mathcal{D}_{macro} = \{(\boldsymbol{\omega}^{(n)}, F^{(n)})\}_{n=1}^{N}$, which records the final fitness scores $F^{(n)}$ under N reward weight configurations $\boldsymbol{\omega}^{(n)}$ and provides task-level evidence on how reward-component weights affect task success. 2) $\mathcal{D}_{micro} = \{(s_0^{(n)}, s_1^{(n)}, \ldots, s_T^{(n)})\}_{n=1}^{N}$, which records per-step transitions among structural variables. Here, $s_t^{(n)} \in \mathbb{R}^d$ captures the instantaneous values of the d physical variables underlying the reward components, instead of the raw, less interpretable observations o_t. The datasets are then aggregated across specifications and source tasks for prior learning. N counts weight configurations per specification, not the total dataset size or the number of evaluation episodes. See Suppl. Sec. S-E for an example.

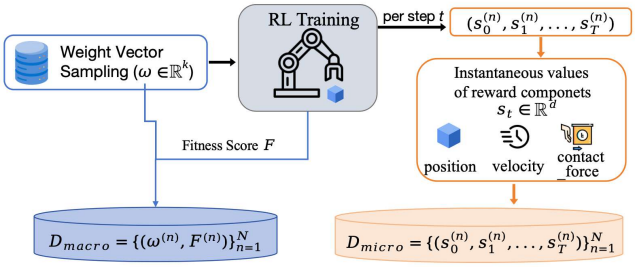


Fig. 3. **Illustration of the Hierarchical Data Collection Pipeline.** Leveraging systematically perturbing reward-component weights $\boldsymbol{\omega} \in \mathbb{R}^k$ and executing RL training, we obtain two levels of interventional data from one collection procedure. **Left (Blue):** The macro-level dataset $\mathcal{D}_{macro}$ records the final fitness scores $F^{(n)}$ evaluated across different weight configurations $\boldsymbol{\omega}^{(n)}$. **Right (Orange):** The micro-level dataset $\mathcal{D}_{micro}$ logs dense, per-step trajectories from policies trained under the sampled configurations. Each state $s_t \in \mathbb{R}^d$ explicitly captures the instantaneous values of the specific physical variables underlying the reward components, such as position, velocity, and contact force.

Structural Prior Extraction. Given $\mathcal{D}_{macro}$, we use a LimiX-based estimator [36] to propose an initial relation-score matrix over the shared reward-component vocabulary. LimiX provides conditional predictions over structured data; causal pre-training alone does not identify physical intervention effects. Specifically, the LimiX-based estimator processes the macro-level reward-component intervention table and assigns continuous scores to candidate directed relations. This yields $M_{raw} \in \mathbb{R}^{d \times d}$, where $M_{raw}^{i,j}$ scores a proposed relation from v^j to v^i, rather than an identified $do(v^j)$ effect on v^i. From M_{raw}, we derive the coarse structural prior $\langle M_{prior}, M_{conf} \rangle$: M_{prior} preserves the directed topology encoded by M_{raw}, including initial edge existence and directions, while $M_{conf} \in [0, 1]^{d \times d}$ encodes normalized effect magnitudes for the subsequent Soft Fusion module. For $i, j \in \{1, \ldots, d\}$, M_{conf} is computed as

$$M_{conf}^{i,j} = \frac{|M_{raw}^{i,j}|}{\max_{1 \le p,q \le d} |M_{raw}^{p,q}| + \epsilon}, \tag{3}$$

where p and q index all entries in M_{raw}, and ϵ ensures numerical stability. Here, confidence denotes a magnitude-based structural weight: M_{conf} measures relative score magnitude, not calibrated uncertainty or an edge-correctness probability, and depends on the estimator's scale. The benefit of this coarse structural prior over unguided purely data-driven discovery is shown in the ablation studies (Section V-E).

Although this process provides useful structural initialization, the coarse structural prior can still contain biased relation scores and structural hallucinations, as illustrated in the bottom-left panel of Fig. 4. These errors arise because the causal foundation model only receives macro-level intervention outcomes, without observing micro-level trajectories that reveal state-dependent transient interactions. Imposing M_{prior} as a hard constraint can therefore bias the refined reward causal prior and degrade subsequent reward generation. Thus, we calibrate this coarse prior with micro-level structural-variable trajectories through the Joint Optimization Module introduced next.

B. Joint Optimization for Reward Causal Prior Refinement

The goal of this stage is to obtain the refined reward causal prior M^*_{inv} by calibrating the coarse structural prior with micro-level trajectories. This stage addresses two issues left by Section IV-A. First, a single static topology can confuse persistent reward-relevant relations with state-dependent transient physical interactions, such as contact-induced effects that appear only under specific physical states. Fig. 4 illustrates this limitation: the coarse structural prior incorrectly retains the transient correlate `ball_velocity` while missing the contact-dependent `impact_penalty` relation. A simplified analysis of this static-topology limitation is provided in Suppl. Sec. S-D. Following the Independent Causal Mechanisms (ICM) principle [34], we therefore model reward-relevant transitions using a learnable causal skeleton together with state-dependent transient interactions. Second, as discussed in Section IV-A, the coarse structural prior M_{prior} may contain structural hallucinations and biased relation scores, so directly enforcing it can bias the refined prior.

To address these coupled challenges, we propose a joint continuous optimization framework that accounts for state-dependent transient physical interactions while calibrating the coarse structural prior. This module optimizes the learnable causal skeleton M_{inv} and obtains its converged form M^*_{inv} as the refined reward causal prior. Prior refinement is driven by the unified objective:

$$\min_{M_{inv},\theta} \mathcal{L} = \mathbb{E}_{(s_{t-1},s_t)\sim\mathcal{D}_{micro}} \Big[\mathcal{L}_{MSE}\Big(s_t, M_{inst}(s_{t-1})s_{t-1}\Big)\Big] + \lambda h(M_{inv}) + \mathcal{L}_{soft}. \tag{4}$$

Here, M_{inv} denotes the learnable causal skeleton whose converged form M^*_{inv} is retained as the reward causal prior, and θ parameterizes the state-dependent transient component introduced by EMD. The reconstruction term compares each transition pair $(s_{t-1}, s_t) \sim \mathcal{D}_{micro}$ with the locally linear prediction $M_{inst}(s_{t-1})s_{t-1}$, where $M_{inst}(s_{t-1})$ is the state-dependent coefficient matrix constructed below. The coefficient λ weights the acyclicity regularization $h(M_{inv})$ (Suppl. Sec. S-B-C), and $\mathcal{L}_{soft}$ denotes the Confidence-Aware Soft Fusion loss for calibrating the coarse structural prior.

Since M_{inv} is retained as the reward-relevant causal skeleton for reward generation, we regularize it as a directed acyclic structure to avoid circular dependencies in the causal prior provided to the LLM. This regularization applies to the reward-structure matrix, rather than the full time-lagged physical dynamics. Multiplicative gating rescales its entries and does not introduce feedback edges or self-loops outside its support. Specifically, we impose the DAGMA acyclicity constraint [37] on M_{inv}:

$$h(M_{inv}) = -\log\det(\alpha I - M_{inv} \odot M_{inv}) + d\log\alpha, \tag{5}$$

where $\alpha > 0$ is the DAGMA scalar hyperparameter, I is the identity matrix, $d = |\mathcal{V}|$ is the number of structural variables, and $\odot$ denotes the Hadamard product. We set $\alpha = 1$; the log-determinant requires $\rho(M_{inv} \odot M_{inv}) < \alpha$, where ρ is the spectral radius. A finite regularization penalty alone does not guarantee exact acyclicity. The effect of this acyclicity regularization is further examined in Suppl. Sec. S-F. The objective in Eq. 4 is implemented by two modules introduced below: EMD constructs the state-dependent coefficient matrix $M_{inst}(s_{t-1})$, while Confidence-Aware Soft Fusion defines $\mathcal{L}_{soft}$ to calibrate the coarse structural prior.

Explicit Mechanism Decoupling (EMD). EMD constructs the state-dependent coefficient matrix $M_{inst}(s_{t-1})$ used in the reconstruction term of Eq. 4. Instead of using a single static topology for all structural-variable transitions, EMD constructs a state-dependent coefficient matrix by combining the learnable causal skeleton with state-dependent transient physical interactions. Specifically, $M_{inv} \in \mathbb{R}^{d\times d}$ denotes the learnable causal skeleton retained for reward generation, and $M_{trans}(s_{t-1};\theta) \in \mathbb{R}^{d\times d}$ denotes the transient component parameterized by an MLP, where s_{t-1} is the previous structural-variable state and θ denotes the MLP parameters. The state-dependent coefficient matrix is constructed as

$$M_{inst}(s_{t-1}) = M_{inv} \odot \Big(\mathbf{1}_{d\times d} + M_{trans}(s_{t-1};\theta)\Big), \tag{6}$$

where $\mathbf{1}_{d\times d}$ is an all-ones matrix, and $\odot$ denotes the Hadamard product. This multiplicative form allows $M_{trans}(s_{t-1};\theta)$ to adjust candidate-edge strengths across physical states while anchoring the optimization to M_{inv}.

By constructing $M_{inst}(s_{t-1})$ at each step, EMD allows $M_{trans}(s_{t-1};\theta)$ to account for short-lived variations during reconstruction, while M_{inv} remains the skeleton retained for reward generation. The coefficient matrix M_{inst} has support contained in M_{inv}. However, the MLP accesses the full state, so its prediction can depend on a variable even when the corresponding matrix entry is zero. The mask therefore defines a reward-structure bottleneck, not the complete functional causal graph. The effect of this decoupled formulation over static causal modeling is analyzed in Suppl. Sec. S-D, and the overall contribution of EMD is evaluated in Section V-E (Causal-ARD w.o. EMD).

Confidence-Aware Soft Fusion. While EMD accounts for state-dependent transient physical interactions in the reconstruction term, a reliable causal skeleton also requires calibrating the coarse structural prior M_{prior}. Using M_{prior} as a rigid structural mask would force the optimization to inherit its structural hallucinations and biased relation scores. We therefore define the Confidence-Aware Soft Fusion loss as

$$\mathcal{L}_{soft} = \gamma \| M_{conf} \odot (M_{inv} - M_{prior}) \|_F^2, \tag{7}$$

where γ controls the loss strength, M_{conf} contains the magnitude-based fusion weights in Eq. 3, and $\| \cdot \|_F$ denotes the Frobenius norm. The setting of γ is discussed in Suppl. Sec. S-B-C.

This term imposes stronger regularization on relations with larger initial scores and permits more adjustment of smaller-score relations. Together with the reconstruction term in Eq. 4, Soft Fusion softly calibrates the coarse structural prior using micro-level structural-variable trajectories. Thus, Soft Fusion permits deviations from M_{prior}; large initial scores need not imply reliable edges. The qualitative changes before and after joint optimization are illustrated in Fig. 4, and the contribution of Soft Fusion is evaluated in Section V-E (Causal-ARD w.o. Soft-Fusion).

We optimize the regularized objective in Eq. 4 with penalty weight $\lambda = 0.1$. This is a penalty formulation, not an augmented Lagrangian with an equality-constraint guarantee. This optimization balances transition reconstruction, acyclicity regularization, and magnitude-weighted prior calibration. Upon convergence, the optimized causal skeleton is extracted as M^*_{inv} and retained as the refined reward causal prior. A visualization of this prior is provided in Suppl. Fig. S-1.

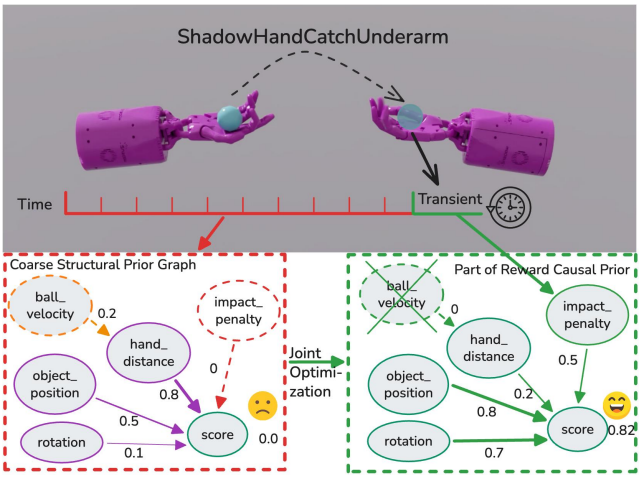


Fig. 4. **Joint Optimization for Reward Causal Prior Refinement.** The top panel shows a physical trajectory in the *ShadowHandCatchUnderarm* task, where ball-hand contact induces a state-dependent transient physical interaction. The bottom-left panel shows the coarse structural prior M_{prior} before joint optimization, which contains biased relation scores, a hallucinated `ball_velocity` node, and a missing contact-related `impact_penalty` relation. The bottom-right panel shows the refined reward causal prior after joint optimization, where the hallucinated node is removed and the contact-related relation is included when supported by transition data. Overall, EMD accounts for state-dependent transient physical interactions during reconstruction, while Confidence-Aware Soft Fusion calibrates the coarse structural prior using observed transitions and relative score magnitudes.

C. Causal Automated Reward Design (Causal-ARD)

After prior refinement, M^*_{inv} must be converted into constraints that can guide executable reward synthesis. Causal-ARD addresses this interface problem by translating the continuous prior matrix into a structured causal topology prompt and grounding its variables in the target task's observation space. The LLM then uses the task objective together with the grounded causal relations to select reward components, remove spurious correlates, and generate the final reward function R^*.

Causal Graph to Prompt Translation. Since LLMs are better suited to structured text than continuous numerical matrices due to tokenization inefficiencies and spatial reasoning limits [38], [39], Causal-ARD employs a deterministic serialization protocol. To select edges for the prompt, we apply one-dimensional 2-means clustering to the absolute learned coefficients $\mathcal{W} = \{|M^{*,i,j}_{inv}|\}^d_{i,j=1}$, where j and i denote the source and target nodes, respectively. This partitions $\mathcal{W}$ into low- and high-magnitude clusters by minimizing intra-cluster variance. We set $\tau = \frac{\mu_{noise}+\mu_{signal}}{2}$ from the cluster centroids. This is a magnitude-based selection heuristic, not a test of causal significance. The selected edges are serialized as *source*, *target*, and *signed coefficient*, as illustrated in Fig. 5. Their values are structural guidance, not reward coefficients or probabilities. The prompt combines these physical-variable relations with $\mathcal{E}_F$, which is retained from the macro-level prior rather than updated by transition reconstruction. Score paths are reward-relevance hypotheses whose utility direction must be grounded in the target objective. The full prompt is provided in Suppl. Sec. S-B-A.

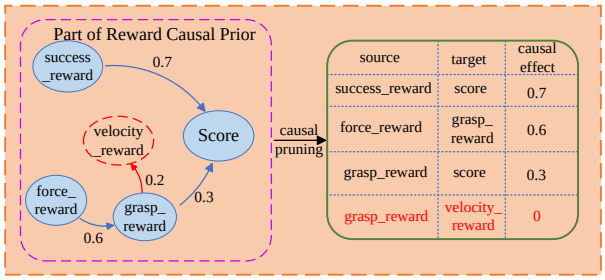


Fig. 5. **Translation of the reward causal prior into explicit topological triplets.** The selected relations from M^*_{inv} are serialized with *source*, *target*, and a signed learned coefficient, labeled as causal effect in the illustration. These triplets are embedded into the causal topology prompt to guide explicit causal pruning during zero-shot reward generation.

Causal Pruning. Conditioned on the *Causal Topology Prompt*, the LLM generates the reward function under explicit causal constraints. The prompt supplies candidate reward-relevant links and excludes components unsupported by the supplied prior. The LLM first matches the serialized nodes to variables available in the target observation space, then retains components consistent with the task objective and the supplied topology. During code generation, this pruning heuristic assigns zero coefficients to excluded terms. A missing path in a finite-data prior does not prove causal irrelevance and can exclude useful shaping terms. As highlighted by the red annotations in Fig. 5, the coefficient of an excluded component is set to zero in the generated code.

Executable Output of Causal-ARD. Ultimately, for an unseen target task, the LLM receives a structured input context consisting of (1) the textual task description, (2) the raw observation space, and (3) the Causal Topology Prompt. Notably, reward components whose required embodiment-specific variables are absent from the observation space, such as finger-related variables in ManiSkill2 or real-world tasks, are not used by the LLM during reward design. The target objective determines component selection, desired direction, and reward sign: for example, door opening and closing require opposite target directions despite sharing physical variables. The LLM organizes the grounded components into a Python reward function in a single pass. By producing executable code for the unseen environment without iterative trial-and-error, this zero-shot process avoids computationally expensive evolutionary RL loops. The generated reward function can be directly executed by the underlying RL algorithm, as validated in Section V-D. The overall procedure, from offline prior learning to executable reward construction, is summarized in Alg. 1.

Algorithm 1 Causal-ARD: Offline Prior Learning and Zero-Shot Reward Construction

Input: An interventional dataset $\mathcal{D}_{off} = \{\mathcal{D}_{macro}, \mathcal{D}_{micro}\}$, a task description $\mathcal{T}$, observation nodes $\mathcal{V}$, a causal foundation model $\mathcal{M}_f$, and an LLM $\mathcal{M}_{LLM}$

Output: A zero-shot reward function represented in Python code R^*

Estimate the coarse structural prior and magnitude weights: $\langle M_{prior}, M_{conf}, \mathcal{E}_F \rangle \leftarrow M_f(\mathcal{V}, \mathcal{D}_{macro})$
Initialize the invariant skeleton $M_{inv} \in \mathbb{R}^{d\times d}$ and the transient modulator parameters θ
while not converged **do**
  Sample transition batches $(s_{t-1}, s_t) \sim \mathcal{D}_{micro}$
  Construct the state-dependent coefficient matrix: $M_{inst}(s_{t-1}) \leftarrow M_{inv} \odot \Big(\mathbf{1}_{d\times d} + M_{trans}(s_{t-1};\theta)\Big)$
  Compute the reconstruction loss: $\mathcal{L}_{MSE} \leftarrow \|s_t - M_{inst}(s_{t-1})s_{t-1}\|_2^2$
  Compute the joint objective:
    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{MSE} + \lambda h(M_{inv}) + \gamma \|M_{conf} \odot (M_{inv} - M_{prior})\|_F^2$
  Update M_{inv} and θ using $\nabla \mathcal{L}_{\text{total}}$
end while
Retain the refined reward causal prior: $M^*_{inv} \leftarrow M_{inv}$
Compute an adaptive threshold τ via 1D 2-means clustering on $\{|M^{*,i,j}_{inv}|\}$
Extract the active causal triplets: $\mathcal{E} \leftarrow \{(v_j, v_i, M^{*,i,j}_{inv}) \mid |M^{*,i,j}_{inv}| \geq \tau\}$
Augment $\mathcal{E}$ with $\mathcal{E}_F$ for the prompt's `score` node
Serialize $\mathcal{E}$ to construct the causal topology prompt $\mathcal{P}_{causal}$
Ground $\mathcal{P}_{causal}$ in the target observation nodes and remove unavailable variables
Construct the reward: $R^* \leftarrow \mathcal{M}_{LLM}(\mathcal{T}, \mathcal{V}, \mathcal{P}_{causal})$
return R^*

V. EXPERIMENTS

In this section, we evaluate Causal-ARD through extensive experiments on zero-shot task performance, target-task reward-search requirements, and generalization across diverse environments and robotic morphologies. We further demonstrate its practical feasibility by zero-shot synthesizing reward functions for real-world robotic tasks and validating their alignment offline. To control for the choice of language model, all LLM-driven methods, including ours and the baselines, use the same foundation model, `deepseek-v3-2-251201` [40] (see Suppl. Sec. S-H-A for results with other open-source LLMs). We also use the pre-trained causal foundation model LimiX [36] across all experiments, with comparisons to alternative causal models provided in Suppl. Sec. S-H-B.

A. Experimental Setup

Dexterity (Isaac Gym) [41]: We evaluate our approach on 20 tasks from the Shadow Hand manipulation suite [3], [41] (detailed in Suppl. Sec. S-C). These tasks feature high-dimensional state-action spaces and dense transient physical interactions. The abundance of redundant kinematic variables provides a testbed for evaluating whether Causal-ARD can prune spurious reward components and construct effective rewards in the zero-shot setting.

ManiSkill2: To validate generalization across diverse environments, we utilize the ManiSkill2 benchmark [42] (detailed in Suppl., Sec. S-C). This suite of robotic manipulation tasks tests whether the reward causal prior can support zero-shot reward construction across varying tasks and robotic morphologies.

B. Baselines

Zero-Shot LLM (ZS-LLM): A purely semantic baseline. Following established protocols [3], we prompt the LLM with raw environment source code and task descriptions to synthesize reward functions in a single pass. This represents foundation model capabilities relying solely on semantic correlations, without explicit causal guidance or environmental feedback.

Eureka [3]: An evolutionary ARD framework utilizing LLMs to heuristically mutate and refine reward functions through computationally intensive trial-and-error interactions and environmental feedback.

URDP [22]: A bi-level optimization framework integrating LLM-driven structural refinement with uncertainty-aware Bayesian optimization. Although highly sample-efficient compared to Eureka, this posterior-correction method fundamentally lacks explicit causal mechanism decoupling.

C. Evaluation Metrics

We evaluate the automated reward design pipelines across two metrics to quantify both task performance and computational footprint. **(i) Success Rate (SR):** The primary indicator of task mastery and policy robustness. To ensure a fair comparison with the baseline methods, the SR for ManiSkill2 tasks is calculated utilizing the last 50% of test episodes

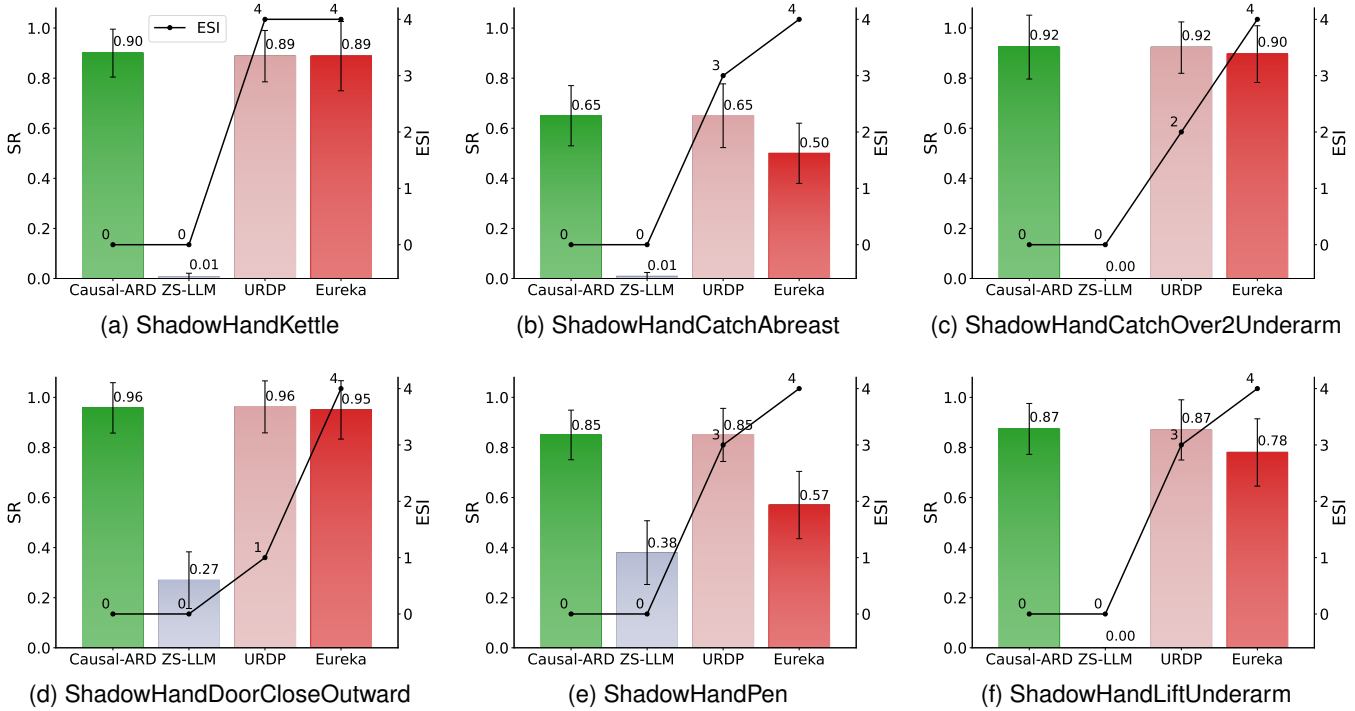


Fig. 6. **Zero-shot task performance across 6 unseen dexterous manipulation tasks.** The bar charts, corresponding to the left y-axis, represent the final SR of the generated reward functions. The solid black line, corresponding to the right y-axis, denotes the ESI required to achieve the reported performance. "Causal-ARD" denotes our full zero-shot reward design pipeline, including offline reward causal prior learning and causal-constrained reward construction. Across all unseen tasks, this pipeline achieves performance comparable to state-of-the-art iterative methods with zero evolutionary iterations (ESI $= 0$), avoiding the costly trial-and-error loops required by Eureka and URDP. In contrast, the unguided ZS-LLM baseline exhibits inferior performance, particularly on dynamic and long-horizon tasks.

from each evaluation rollout, whereas full test results are aggregated for the Dexterity suite. **(ii) Evolutionary Search Iterations (ESI):** The number of feedback-driven reward refinement rounds after the initial generation batch. For zero-shot methods, all candidate reward functions are generated before environment evaluation. Evaluating this fixed candidate set does not increase ESI because its results do not trigger reward revision. ESI excludes offline prior learning and fixed-candidate policy training and evaluation; ESI $= 0$ does not imply zero target-environment interaction.

D. Results

To evaluate Causal-ARD, we conduct experiments on 20 Dexterity continuous control tasks. We use a 14/6 task split, where 14 tasks provide offline data for learning the reward causal prior and the remaining 6 unseen tasks form the zero-shot evaluation suite. This split tests whether reward-relevant structure learned from source tasks can support reward construction on held-out tasks without task-specific feedback. On these unseen tasks, we compare our full zero-shot reward generation pipeline against the semantic baseline ZS-LLM and iterative ARD frameworks, Eureka and URDP. During evaluation, each round prompts the LLM to generate 16 candidate reward functions in parallel. Both Causal-ARD and ZS-LLM operate in the zero-shot setting (ESI $= 0$), generating a fixed candidate set without feedback-driven revision, whereas the reported ESI for Eureka and URDP denotes the evolutionary iterations required to reach their converged SR. The offline cost of learning the reward causal prior is reported separately in Suppl. Sec. S-B-D.

Fig. 6 compares task performance and computational cost across the 6 unseen tasks using a dual-axis design. Bars on the left y-axis report the final SR (↑) achieved by the generated rewards, while lines on the right y-axis indicate the ESI (↓) required to reach that performance. Causal-ARD achieves state-of-the-art-level task performance in a single reward-generation pass with no evolutionary iterations (ESI $= 0$). This comparison yields three key insights as follows.

Causal Priors Improve Zero-Shot LLM Reward Generation. The lower SR of ZS-LLM indicates a benefit from the additional offline structural guidance in these tasks. Without this guidance, ZS-LLM can introduce plausible but poorly aligned reward terms, leading to near-zero SRs on challenging unseen tasks such as *ShadowHandLiftUnderarm* and *ShadowHand-Kettle*. In contrast, Causal-ARD uses the learned reward causal prior to prune spurious components before policy training, providing structural guidance for zero-shot reward generation.

SOTA-Level Zero-Shot Performance. Across the six unseen dexterous manipulation tasks, Causal-ARD achieves performance comparable to iterative ARD baselines. For example, it attains SRs of 0.92 and 0.87 on the challenging *Catch Over 2 Underarm* and *Lift Underarm* tasks, respectively. These results outperform the multi-iteration Eureka baseline and match URDP, which requires up to 3 evolutionary search iterations on these two tasks. Notably, this is achieved in a

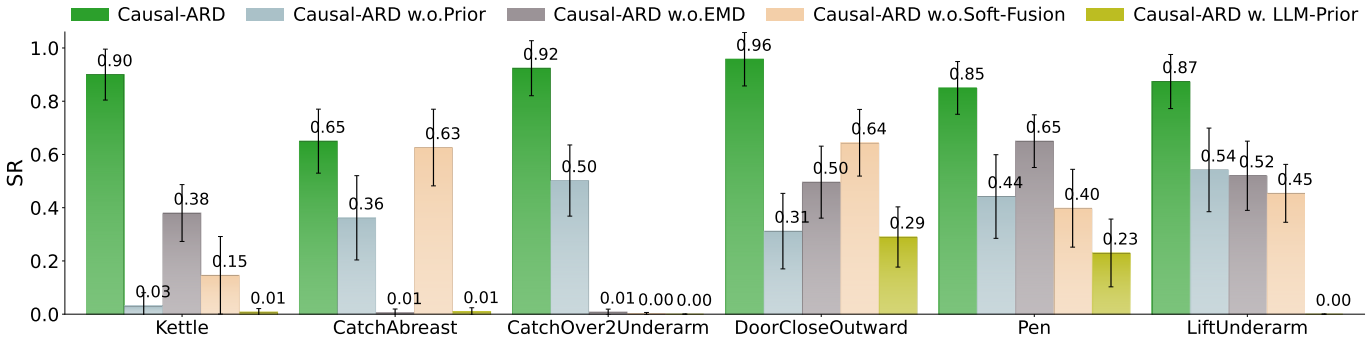


Fig. 7. **Ablation studies across 6 unseen zero-shot tasks.** Performance drops when key Causal-ARD components are removed or when the learned reward causal prior is replaced by an LLM-inferred semantic prior.

zero-shot setting, without feedback-driven reward refinement or additional generation rounds.

Zero-Shot Efficiency on Unseen Tasks. Fig. 6 measures deployment-time search rounds, rather than end-to-end computational cost. Iterative ARD baselines require repeated evolutionary search to reach their converged SR, with URDP requiring 1–4 ESIs and Eureka requiring 4 ESIs across the evaluated tasks. Each ESI trains and evaluates candidate reward functions through large-scale RL environment interactions, which is time-consuming in continuous control tasks and takes approximately one hour on eight NVIDIA A800 GPUs in our setting. In contrast, Causal-ARD achieves competitive task performance with ESI $= 0$, generating executable rewards for unseen tasks without feedback-driven refinement. This comparison does not establish lower total cost once offline prior learning is included.

E. Ablation Experiments

To analyze the contribution of each component in Causal-ARD, we evaluate four ablated variants across the unseen zero-shot tasks (Fig. 7). Each variant is used within the same Causal-ARD pipeline to measure its impact on task performance: 1) **Causal-ARD w.o. Prior**: removes the coarse structural prior, relying only on offline transition data. 2) **Causal-ARD w.o. EMD**: removes the state-dependent interaction term, reducing prior refinement to a single static structure. 3) **Causal-ARD w.o. Soft-Fusion**: replaces the soft regularization with a rigid structural mask. 4) **Causal-ARD w. LLM-Prior**: retains the same reward construction stage but replaces the learned reward causal prior with a semantic topology inferred by the LLM from the target task description and observation space.

Benefit of the Structural Prior. The *Causal-ARD w.o. Prior* variant consistently underperforms across all unseen tasks, indicating that offline transition data alone is insufficient for stable structure estimation. These results support the usefulness of structural guidance within this pipeline.

Effect of EMD. The *Causal-ARD w.o. EMD* variant has lower SR across the evaluated tasks. Without EMD, the model relies on a single static structure to explain all transitions, reducing its ability to capture data variations. This supports the usefulness of state-dependent modulation for reward-prior refinement.

Soft Fusion Enables Adaptive Use of the Prior. When the structural prior is imposed as a rigid constraint, the *Causal-ARD w.o. Soft-Fusion* variant shows clear performance degradation across multiple tasks. In this setting, the model becomes less adaptive to observed transitions, leading to suboptimal structure estimation. These results indicate that Soft Fusion enables flexible integration between the prior structure and data, which is important for strong performance.

Comparison with LLM-Inferred Priors. The *Causal-ARD w. LLM-Prior* variant performs substantially worse than the full Causal-ARD framework across unseen tasks. Although it uses the same Causal-ARD reward generation procedure, its causal topology is inferred only from the target task description and observation space, without offline interventional data or micro-level trajectories. Thus, the inferred semantic topology can be incomplete or misleading, causing causal pruning to remove useful reward terms or retain spurious correlations. This supports the benefit of the learned prior over the tested semantic topology. These component ablations do not uniquely attribute the gain to correct causal directions rather than component selection or other structural guidance.

F. Cross-Embodiment and Cross-Environment Generalization

We evaluate the cross-embodiment and cross-environment generalization ability of Causal-ARD. Unlike the 14/6 split for unseen-task evaluation within Dexterity, this experiment uses a reward causal prior learned from the full 20-task Dexterity suite. The goal is to examine whether structural variables and causal relations learned from Dexterity can transfer to a different manipulation benchmark. Specifically, we apply Causal-ARD to three ManiSkill2 manipulation tasks whose reward-related variables are covered by the learned structural-variable vocabulary, as detailed under *Compositional Coverage* in Suppl. Sec. S-B-B. This setting introduces a clear domain shift from a high-DoF multi-fingered ShadowHand to single-arm manipulators in a different physics engine.

Causal-ARD constructs rewards before target-environment evaluation, without feedback-driven reward adaptation. As shown in Fig. 8, it achieves strong zero-shot performance (ESI $= 0$) across all tasks, with success rates of 0.89 on

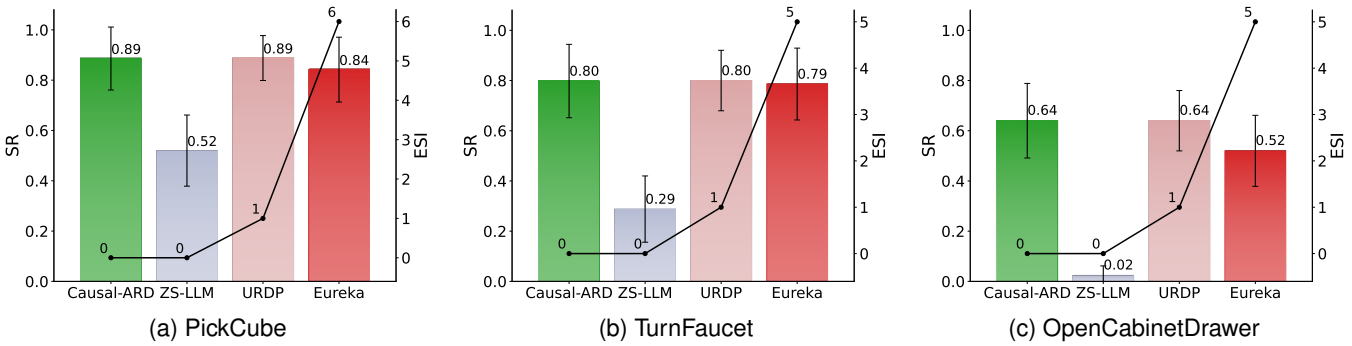


Fig. 8. **Cross-domain generalization in ManiSkill2.** Causal-ARD is applied to three unseen tasks with different robot embodiments and physics environments. It achieves strong success rates in a zero-shot setting, without feedback-driven reward refinement.

PickCube, 0.80 on TurnFaucet, and 0.64 on OpenCabinet-Drawer. In comparison, iterative methods such as Eureka require multiple rounds of environment interaction, up to 6 ESIs, to reach similar performance and still remain less consistent.

These results indicate that the learned reward causal prior transfers across tasks and robot embodiments when the target variables are covered, enabling effective reward construction under substantial domain shifts without feedback-driven reward refinement.

G. Offline Reward Alignment on Real-World Trajectories

We evaluate whether rewards generated by Causal-ARD distinguish successful from failed real-world trajectories. This experiment focuses on reward alignment rather than downstream policy optimization. We use the CASBOT W1 mobile manipulator [43] and adopt *Offline Trajectory Evaluation* [44] to avoid dynamics gaps and hardware risks from direct sim-to-real deployment. Human-teleoperated kinematic trajectories are fed into the generated reward functions to assess alignment independently of RL optimization and sim-to-real discrepancies.

To quantify zero-shot alignment, we collect a balanced human-teleoperated trajectory dataset across three tasks: PickCube, LiftUnderarm, and Scissors (Fig. 9). Each task contains 25 successful and 25 failed trajectories, yielding 625 successful-failed pairs. These pairs share trajectories and are not 625 independent samples. Since reward functions differ in scale, their absolute values are not directly comparable. We evaluate alignment using *Pairwise Distinguishability Accuracy*, equivalent to the Area Under the ROC Curve (AUC) [45]. For each successful-failed pair, a well-aligned reward function should assign a higher time-normalized mean reward $\bar{R}(\tau)$ to the successful trajectory.

As shown in Table I, the ZS-LLM baseline achieves limited distinguishability (average AUC $\approx$ 53.9%). In contrast, reward functions generated by Causal-ARD achieve high alignment accuracy across all tasks (average AUC $\approx$ 96.5%). These results support offline reward alignment on the recorded trajectories; they do not establish closed-loop policy learning or successful sim-to-real deployment.

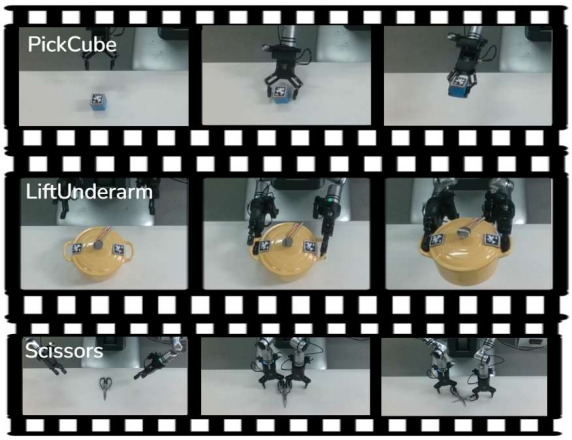


Fig. 9. **Real-world manipulation tasks.** PickCube: picking up a black cube. LiftUnderarm: lifting a pot. Scissors: open scissors.

TABLE I
ZERO-SHOT REWARD ALIGNMENT ACCURACY (AUC) ON REAL-WORLD TRAJECTORIES.

Method	PickCube	LiftUnderarm	Scissors
ZS-LLM (AUC %↑)	62.5%	47.9%	51.3%
Causal-ARD (AUC %↑)	**99.6%**	**95.3%**	**94.5%**

VI. DISCUSSION

Fig. 1 illustrates Optimization Collapse and Specification Gaming in simulated training runs. Table II summarizes the observed baseline failure frequencies.

Table II reports the frequency of these issues across reward functions generated by baselines (Eureka and URDP) over 20 dexterous manipulation tasks. These descriptive baseline statistics do not provide a matched failure-rate comparison with Causal-ARD.

Optimization Collapse: A considerable portion of baseline rewards (21.2% for Eureka and 12.6% for URDP) exhibit cumulative reward and SR that remain near zero during training. Examples are shown in the blue boxes of Suppl. Fig. S-6.

Specification Gaming: A non-negligible fraction of baseline rewards (10.4% for Eureka and 4.7% for URDP) show high reward values with low task success, a pattern consistent

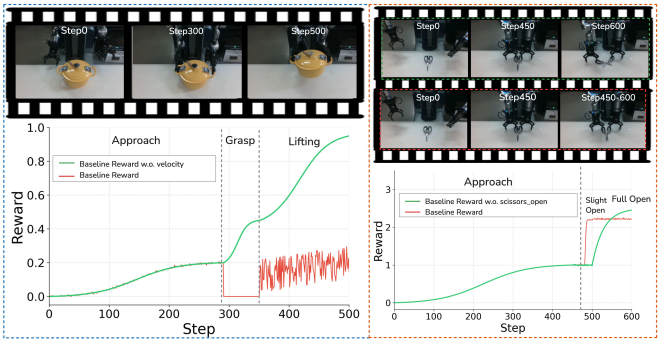


Fig. 10. **Reward signals on recorded real-world trajectories.** Top (blue box): the baseline signal (red) is relatively flat, while the Causal-ARD signal (green) varies with task progress. Bottom (orange box): the baseline signal saturates before task completion, whereas the Causal-ARD signal better reflects progress. No policy optimization is performed on these trajectories.

with misaligned reward optimization (orange boxes in Suppl. Fig. S-6). The displayed Causal-ARD runs show closer agreement between reward and SR. Fig. 10 separately illustrates reward signals along real-world recorded trajectories. Flat or prematurely saturated signals can reveal misalignment, but these offline curves alone do not demonstrate an optimization failure.

TABLE II
STATISTICAL PREVALENCE OF REWARD FAILURE MODES. THE TABLE REPORTS THE PERCENTAGE OF GENERATED REWARD FUNCTIONS THAT FALL INTO OPTIMIZATION COLLAPSE AND SPECIFICATION GAMING ACROSS ALL 20 DEXTERITY TASKS FOR THE BASELINE ARD PARADIGMS (EUREKA AND URDP).

Method	Optimization Collapse (%) ↓	Specification Gaming (%) ↓
Eureka	21.2%	10.4%
URDP	12.6%	4.7%

VII. CONCLUSION

In this work, we present Causal-ARD, a framework for zero-shot automated reward design. Instead of relying on iterative reward refinement, Causal-ARD learns a reusable reward causal prior from offline multi-task data and uses it to constrain LLM-based reward construction. By combining a reward-relevant structural prior with state-dependent modulation, Causal-ARD selects reward components and synthesizes executable rewards in a single pass. Experiments across unseen simulated tasks and different embodiments show strong task performance without deployment-time evolutionary search; offline evaluation on real-world trajectories supports reward alignment. These results support reusing offline structural guidance to reduce feedback-driven reward search on new tasks.

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